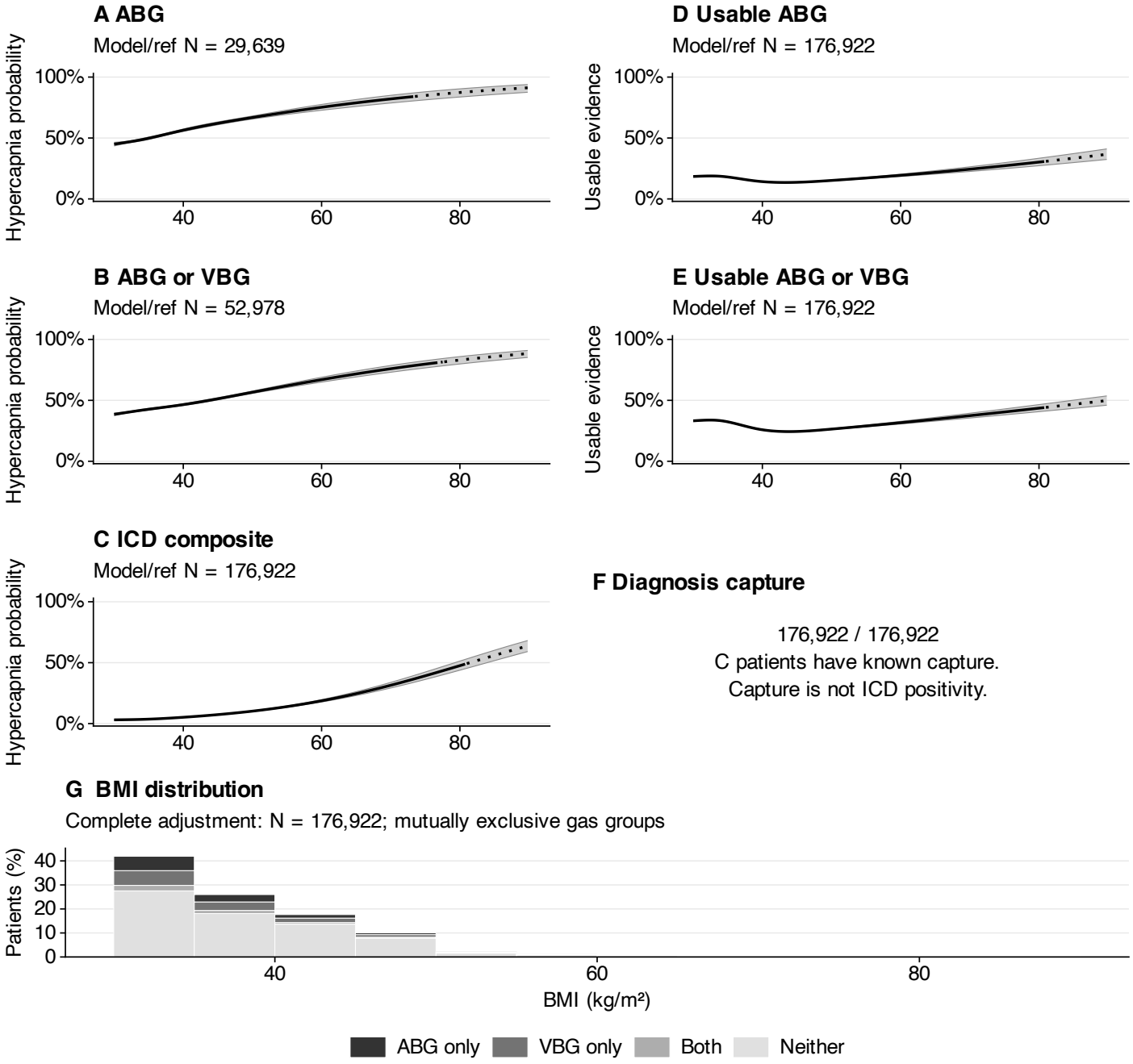
BMI and documented hypercapnia

Non-OBESITY-supported entry | encounter-wide

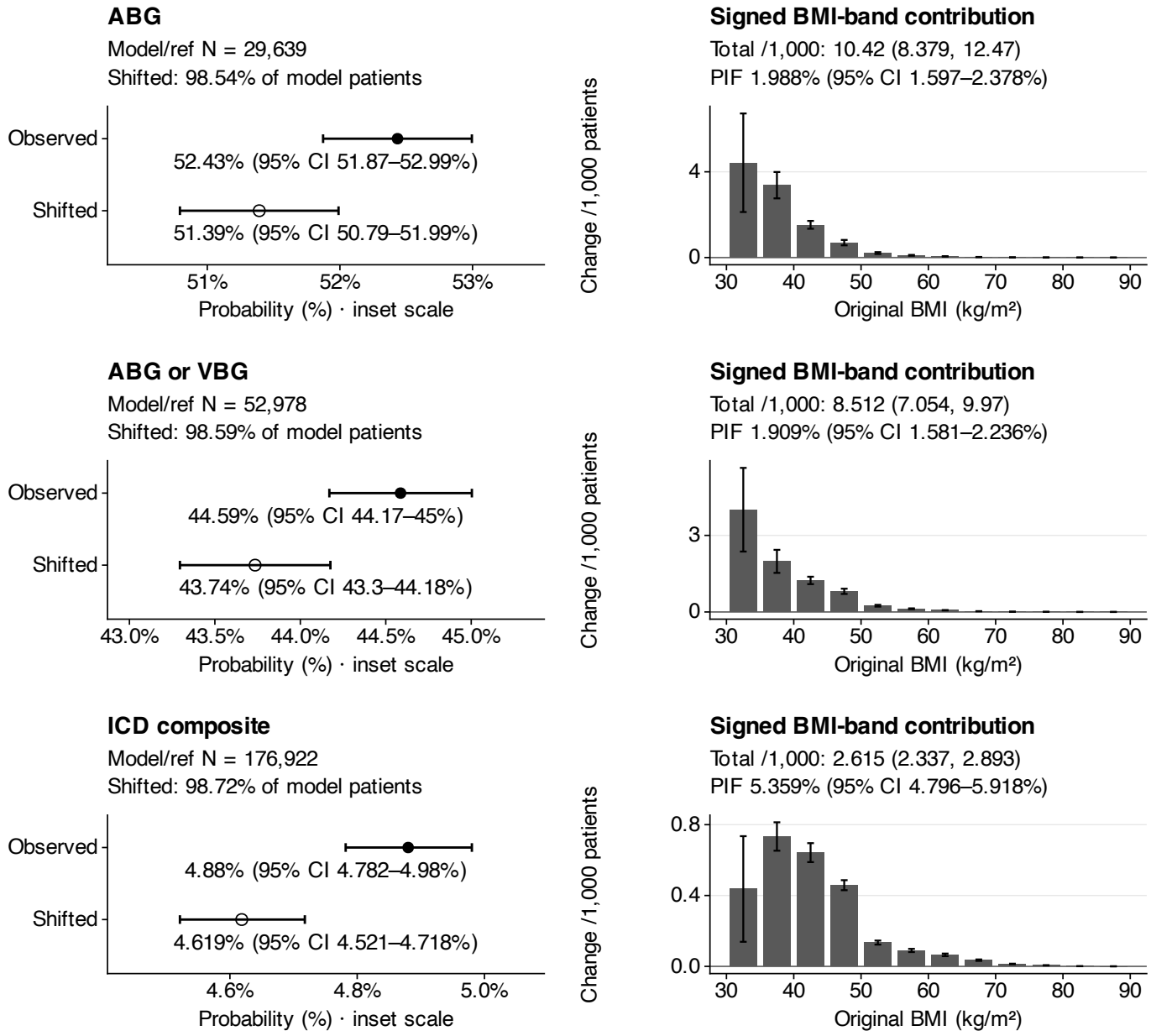


Shading: 95% CI | Dotted: locally sparse | Dashed: outside training BMI range

Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

Modeled probability under a one-unit BMI reduction

Observed BMI >30 decreases by one; BMI =30 is unchanged
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Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired one-unit BMI scenario. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with exact model-based 95% CIs and printed limits. Left probability axes use different labeled inset scales; compare printed values across outcomes. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.